called the *space of k-boundaries* of K. The boundary operators satisfy the fundamental following property:

$$\partial_{k-1} \circ \partial_k \equiv 0 \quad \text{for any} \quad k \geq 1.$$

In other words, any k-boundary is a k-cycle, i.e. $B_k(K) \subseteq Z_k(K) \subseteq C_k(K)$. These notions are illustrated on Figure 6.

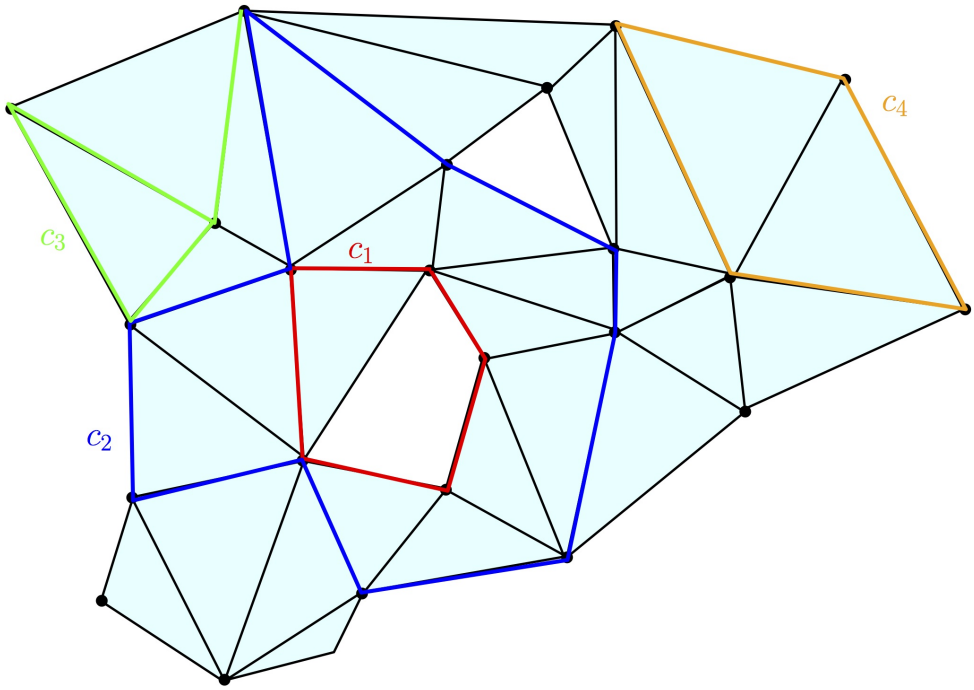

Figure 6: Some examples of chains, cycles and boundaries on a 2-dimensional complex K: c_1, c_2 and c_4 are 1-cycles; c_3 si a 1-chain but not a 1-cycle; c_4 is the 1-boundary, namely the boundary of the 2-chain obtained as the sum of the two triangles surrounded by c_4; The cycles c_1 and c_2 span the same element in $H_1(K)$ as their difference is the 2-chain represented by the union of the triangles surrounded by the union of c_1 and c_2.

Definition 6 (Simplicial homology group and Betti numbers)**.** *The kth (simplicial) homology group of K is the quotient vector space*

$$H_k(K) = Z_k(K)/B_k(K).$$

The kth Betti number of K is the dimension $\beta_k(K) = \dim H_k(K)$ of the vector space $H_k(K)$.

Two cycles $c, c' \in Z_k(K)$ are said to be *homologous* if they differ by a boundary, i.e. is there exists a $(k+1)$-chain d such that $c' = c + \partial_{k+1}(d)$. Two such cycles give rise to the same element of H_k. In other words, the elements of $H_k(K)$ are the equivalence classes of homologous cycles.

Simplicial homology groups and Betti numbers are topological invariants: if K, K' are two simplicial complexes whose geometric realizations are homotopy equivalent, then their homology groups are isomorphic and their Betti numbers are the same.

Singular homology is another notion of homology that allows to consider larger classes of topological spaces. It is defined for any topological space X similarly to simplicial homology except that the notion of simplex is replaced by the notion of *singular simplex* which is just any continuous map $\sigma : \Delta_k \to X$ where Δ_k is the standard k-dimensional simplex. The space of k-chains is the vector space spanned by the k-dimensional singular simplices and the boundary of a simplex σ is defined as the (alternated) sum of the restriction of σ to the $(k-1)$-dimensional faces of Δ_k. A remarkable fact about singular homology it that it coincides with simplicial homology

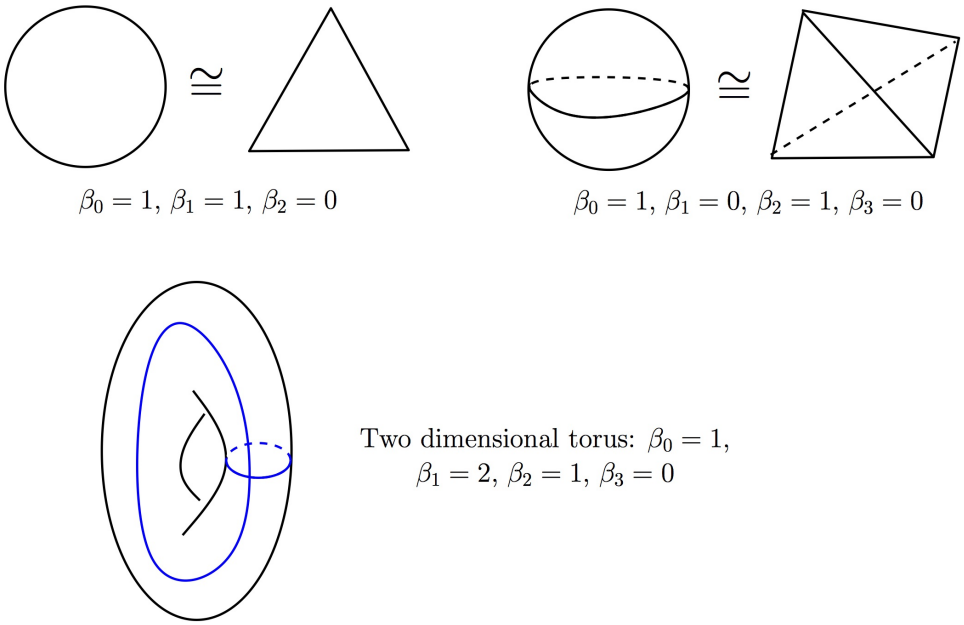

Figure 7: The Betti numbers of the circle (top left), the 2-dimensional sphere (top right) and the 2-dimensional torus (bottom). The blue curves on the torus represent two independent cycles whose homology class is a basis of its 1-dimensional homology group.

whenever X is homeomorphic to the geometric realization of a simplicial complex. This allows us, in the sequel of this paper, to indifferently talk about simplicial or singular homology for topological spaces and simplicial complexes.

Observing, that if $f : X \to Y$ is a continuous map, then for any singular simplex $\sigma : \Delta_k \to X$ in X, $f \circ \sigma : \Delta_k \to Y$ is a singular simplex in Y, one easily deduces that continuous maps between topological spaces canonically induce homomorphisms between their homology groups. In particular, if f is an homeomorphism or an homotopy equivalence, then it induces an isomorphism between $H_k(X)$ and $H_k(Y)$ for any non negative integer k. As an example, it follows from the Nerve Theorem that for any set of points $X \subset \mathbb{R}^d$ and any $r > 0$ the r-offset X^r and the Čech complex $\mathrm{Cech}_r(X)$ have isomorphic homology groups and the same Betti numbers.

As a consequence, the Reconstruction Theorem 2 leads to the following result on the estimation of Betti numbers.

Theorem 3. *Let $M \subset \mathbb{R}^d$ be a compact set such that $\mathrm{reach}_\alpha(d_M) \geq R > 0$ for some $\alpha \in (0,1)$ and let $\mathbb{X}$ be a finite set of point such that $d_H(M, \mathbb{X}) = \varepsilon < \frac{R}{5+4/\alpha^2}$. Then, for every $r \in [4\varepsilon/\alpha^2, R - 3\varepsilon]$ and every $\eta \in (0, R)$, the Betti numbers of $\mathrm{Cech}_r(\mathbb{X})$ are the same as the ones of M^η.*
In particular, if M is a smooth m-dimensional submanifold of $\mathbb{R}^d$, then $\beta_k(\mathrm{Cech}_r(\mathbb{X})) = \beta_k(M)$ for any $k = 0, \cdots, m$.

From a practical perspective, this result raises three difficulties: first, the regularity assumption involving the α-reach of M may be too restrictive; second, the computation of the nerve of an union of balls requires they use of a tricky predicate testing the emptiness of a finite union of balls; third the estimation of the Betti numbers relies on the scale parameter r whose choice may be a problem.

To overcome these issues, Chazal and Oudot (2008) establishes the following result that offers a solution to the two first problems.

Theorem 4. *Let $\mathbb{M} \subset \mathbb{R}^d$ be a compact set such that $\mathrm{wfs}(M) = \mathrm{wfs}_{d_M}(0) \geq R > 0$ and let $\mathbb{X}$ be a finite set of point such that $d_H(M, \mathbb{X}) = \varepsilon < \frac{1}{9}\,\mathrm{wfs}(M)$. Then for any $r \in [2\varepsilon, \frac{1}{4}(\mathrm{wfs}(M) - \varepsilon)]$*

and any $\eta \in (0, R)$,

$$\beta_k(X^\eta) = \mathrm{rk}\left(H_k(\mathrm{Rips}_r(\mathbb{X})) \to H_k(\mathrm{Rips}_{4r}(\mathbb{X}))\right)$$

where rk$(H_k(\mathrm{Rips}_r(\mathbb{X})) \to H_k(\mathrm{Rips}_{4r}(\mathbb{X})))$ *denotes the rank of the homomorphism induced by the (continuous) canonical inclusion* $\mathrm{Rips}_r(\mathbb{X}) \hookrightarrow \mathrm{Rips}_{4r}(\mathbb{X})$.

Although this result leaves the question of the choice of the scale parameter r open, it is proven in Chazal and Oudot (2008) that a multiscale strategy whose description is beyond the scope of this paper provides some help to identify the relevant scales at which Theorem 4 can be applied.

4.3 Statistical aspects of Homology inference

According to the stability results presented in the previous section, a statistical approach to topological inference is strongly related to the problem of *distribution support estimation* and *level sets estimation* under the Hausdorff metric. A large number of methods and results are available for estimating the support of a distribution in statistics. For instance, the Devroye and Wise estimator (Devroye and Wise, 1980) defined on a sample $\mathbb{X}_n$ is also a particular offset of $\mathbb{X}_n$. The convergence rates of both $\mathbb{X}_n$ and the Devroye and Wise estimator to the support of the distribution for the Hausdorff distance is studied in Cuevas and Rodríguez-Casal (2004) in $\mathbb{R}^d$. More recently, the minimax rates of convergence of manifold estimation for the Hausdorff metric, which is particularly relevant for topological inference, has been studied in Genovese et al. (2012). There is also a large literature about level sets estimation in various metrics (see for instance Cadre, 2006; Polonik, 1995; Tsybakov et al., 1997) and more particularly for the Hausdorff metric in Chen et al. (2015). All these works about support and level sets estimation shine light on the statistical analysis of topological inference procedures.

In the paper Niyogi et al. (2008), it is shown that the homotopy type of Riemannian manifolds with reach larger than a given constant can be recovered with high probability from offsets of a sample on (or close to) the manifold. This paper was probably the first attempt to consider the topological inference problem in terms of probability. The result of Niyogi et al. (2008) is derived from a retract contraction argument and on tight bounds over the packing number of the manifold in order to control the Hausdorff distance between the manifold and the observed point cloud. The homology inference in the noisy case, in the sense the distribution of the observation is concentrated around the manifold, is also studied in Niyogi et al. (2008, 2011). The assumption that the geometric object is a smooth Riemannian manifold is only used in the paper to control in probability the Hausdorff distance between the sample and the manifold, and is not actually necessary for the "topological part" of the result. Regarding the topological results, these are similar to those of Chazal et al. (2009d); Chazal and Lieutier (2008b) in the particular framework of Riemannian manifolds. Starting from the result of Niyogi et al. (2008), the minimax rates of convergence of the homology type have been studied by Balakrishna et al. (2012) under various models, for Riemannian manifolds with reach larger than a constant. In contrast, a statistical version of Chazal et al. (2009d) has not yet been proposed.

More recently, following the ideas of Niyogi et al. (2008), Bobrowski et al. (2014) have proposed a robust homology estimator for the level sets of both density and regression functions, by considering the inclusion map between nested pairs of estimated level sets (in the spirit of Theorem 4 above) obtained with a plug-in approach from a kernel estimator.

4.4 Going beyond Hausdorff distance : distance to measure

It is well known that distance-based methods in TDA may fail completely in the presence of outliers. Indeed, adding even a single outlier to the point cloud can change the distance function dramatically, see Figure 8 for an illustration. To answer this drawback, Chazal et al. (2011b) have introduced an alternative distance function which is robust to noise, the *distance-to-a-measure*.